Normal Distributions and Rescaling

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Normal Distribution

Normal Distribution

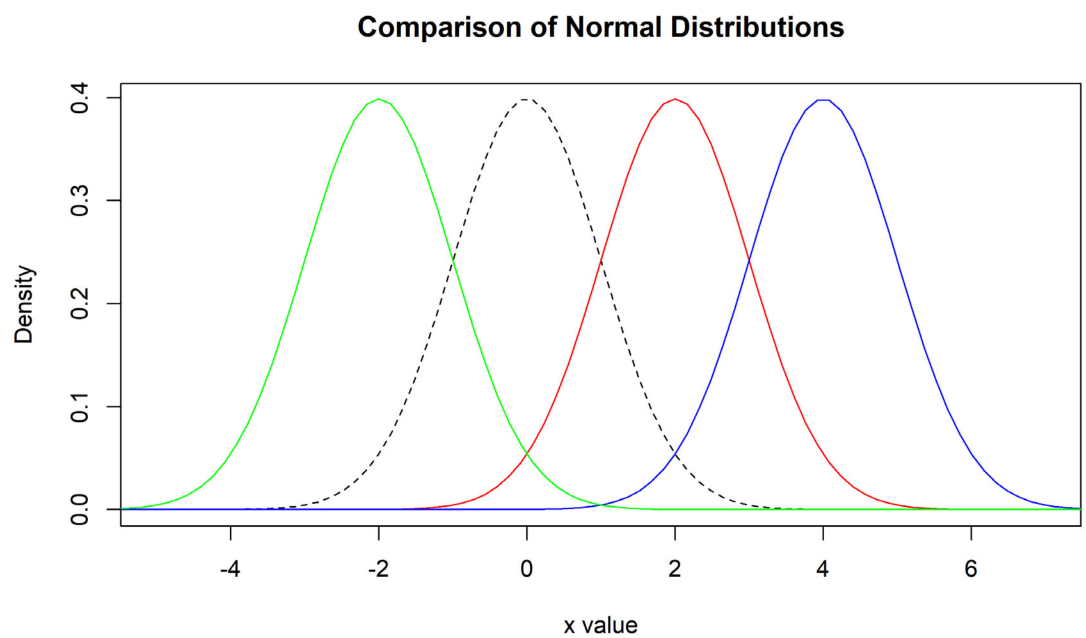
Def: a particular bell-shaped curve that has the following mathematical properties

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$$

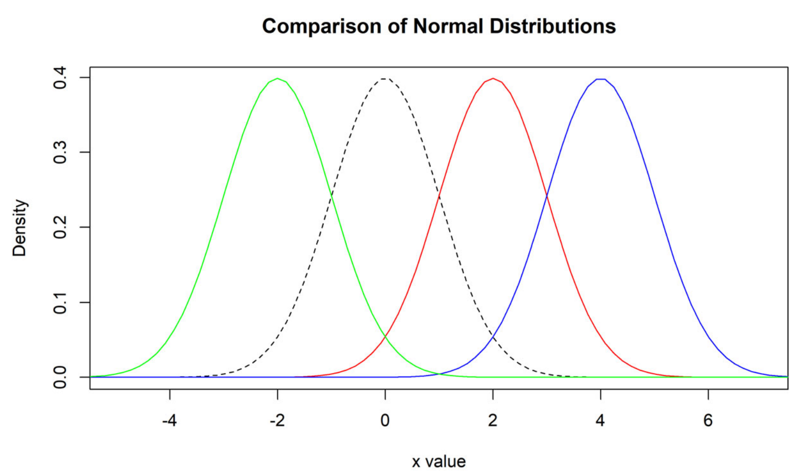
- Formula has two parameters
  - μ
  - σ
- The standard normal $(\mu = 0; \sigma = 1)$ simplifies the equation

Standard Normal with multiple means

- The mean is located at the center of the symmetric curve and is the same as the median.
- Changing μ without changing σ moves the Normal curve along the horizontal axis without changing its variability.



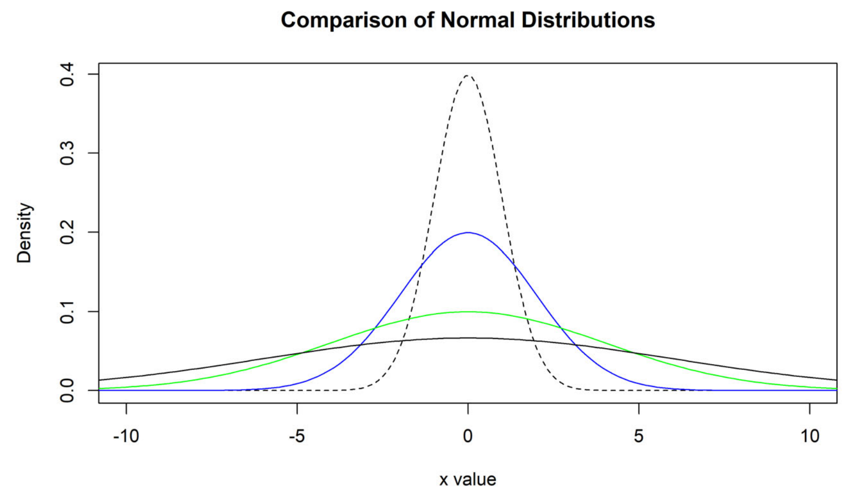
```r
#### Normal Distribution

# Display the normal distributions with various means

x <- seq(-80, 80, length=1000)
hx <- dnorm(x)
colors <- c("red", "blue","green", "green", "gold", "black")

plot(x, hx, type="l", lty=2, xlab="x value",
         ylab="Density", main="Comparison of Normal Distributions",xlim=c(-5, 7))
location<-c(2,4,-2)
for (i in 1:3){
    lines(x, dnorm(x,mean=location[i]), lwd=1, col=colors[i])
}
```



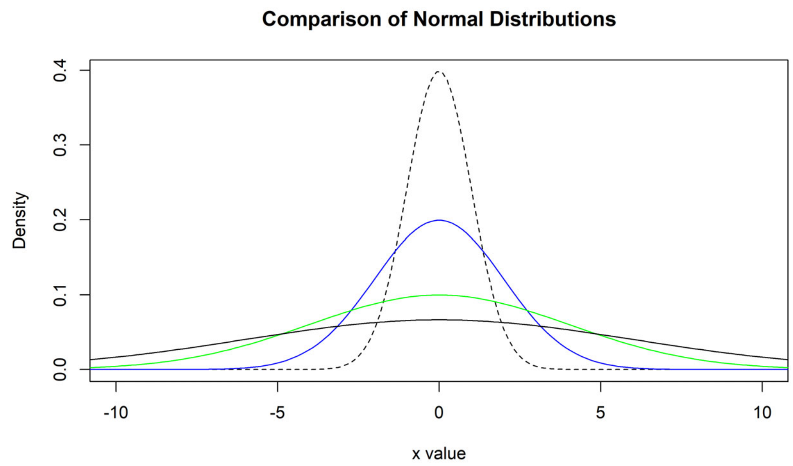
Methods in Psychological Research

Plots of the Standard Normal with multiple standard deviations

- The standard deviation σ controls the variability of a Normal curve.
- When the standard deviation is larger, the area under the normal curve is less concentrated about the mean.
- The standard deviation is the distance from the center to the change-of-curvature points on either side.

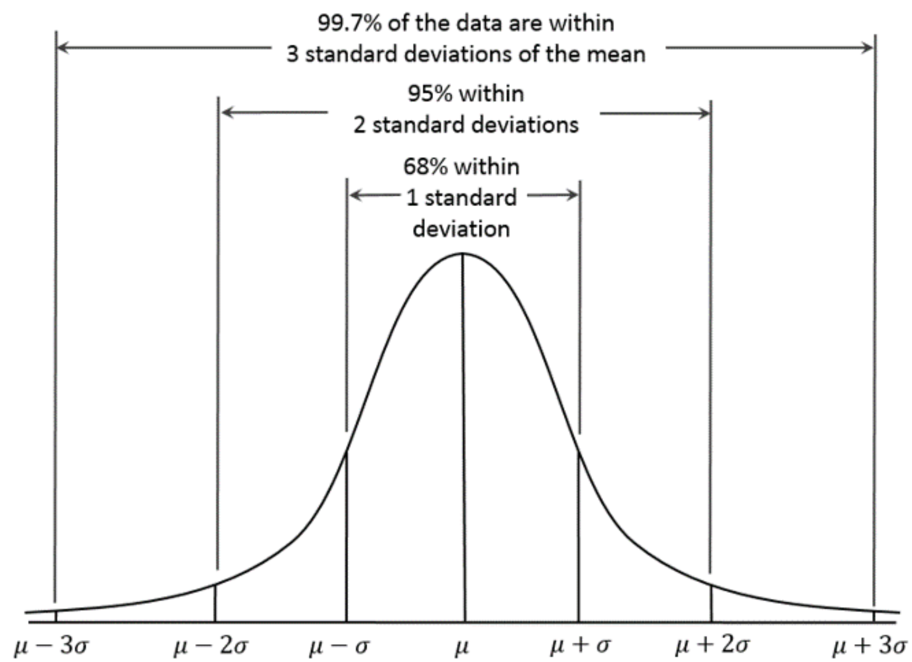


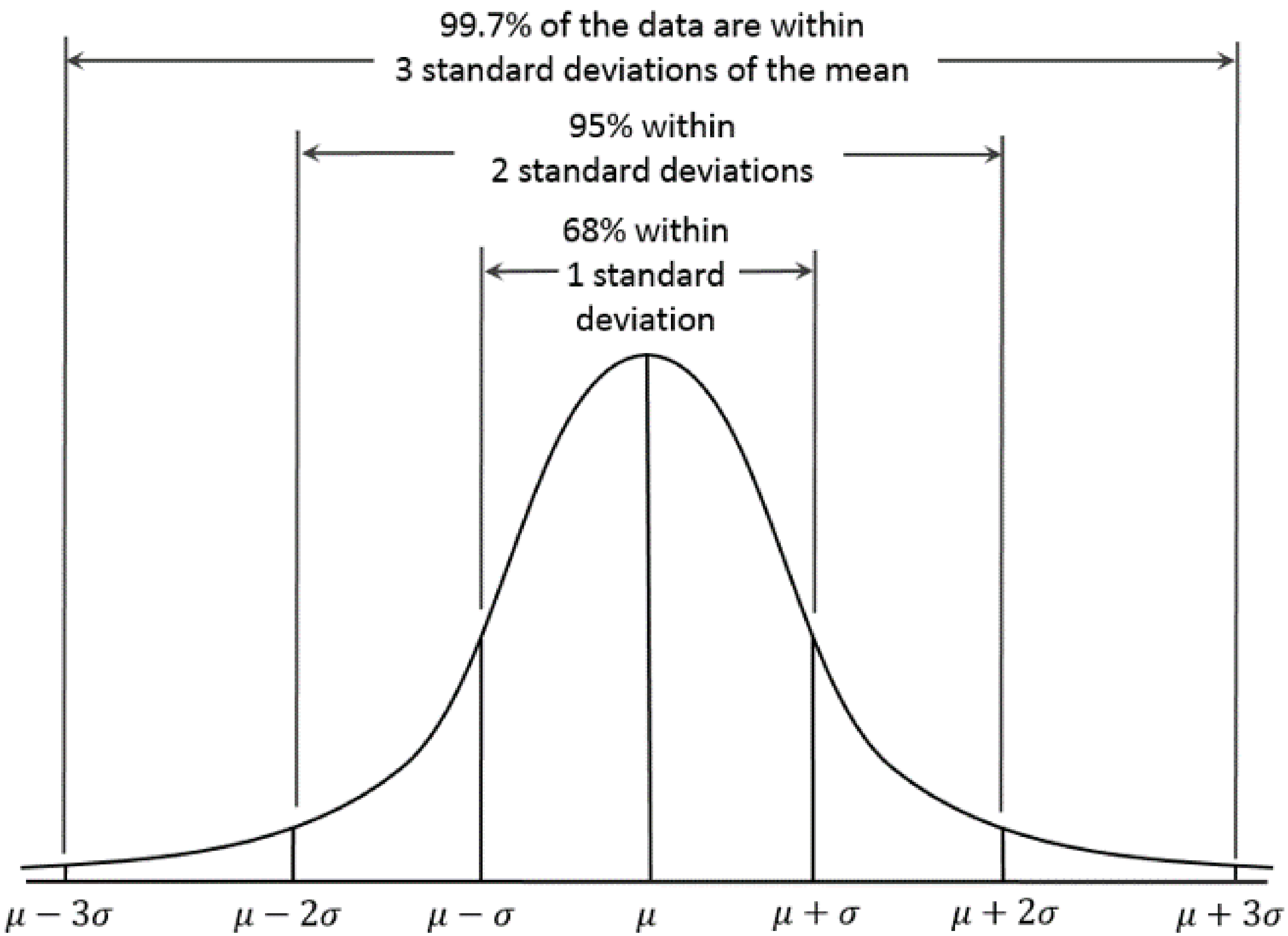
```r
# Display the normal distributions with various standard deviations
plot(x, hx, type="l", lty=2, xlab="x value",
         ylab = "Density", main="Comparison of Normal Distributions",xlim=c(-10, 10))

for (i in c(.5,2,4,6)){
    lines(x, dnorm(x,sd=i), lwd=1, col=colors[i])
}
```



Normal Distribution

- In the Normal distribution, with mean μ and standard deviation σ:
  - approximately 68% of the observations fall within 1 σ of μ
  - approximately 95% of the observations fall within 2 σ of μ
  - approximately 99.7% of the observations fall within 3 σ of μ
- This property is sometimes called: The `68-95-99.7 Rule`

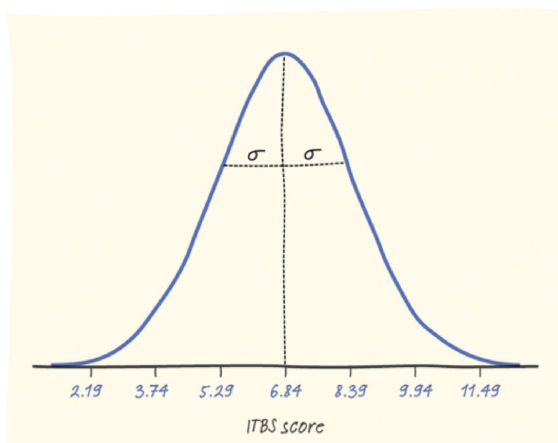


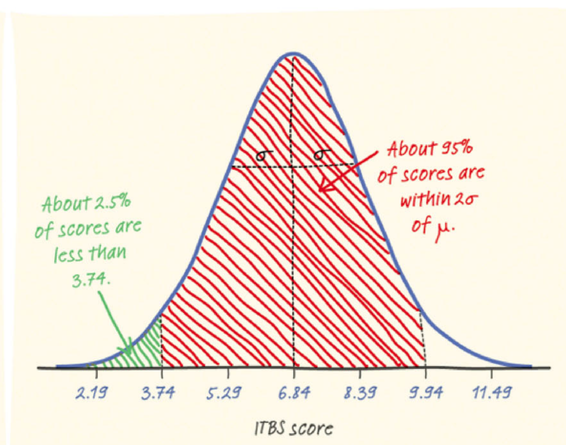

99.7% of the data are within
3 standard deviations of the mean
95% within
2 standard deviations
68% within
1 standard
deviation
$\mu - 3\sigma$
$\mu - 2\sigma$
$\mu - \sigma$
μ
$\mu + \sigma$
$\mu + 2\sigma$
$\mu + 3\sigma$

Worked Example

- The distribution of Iowa Test of Basic Skills (ITBS) vocabulary scores for seventh-grade students in Gary, Indiana, is close to Normal.
- Suppose the distribution is `N(6.84, 1.55)`.

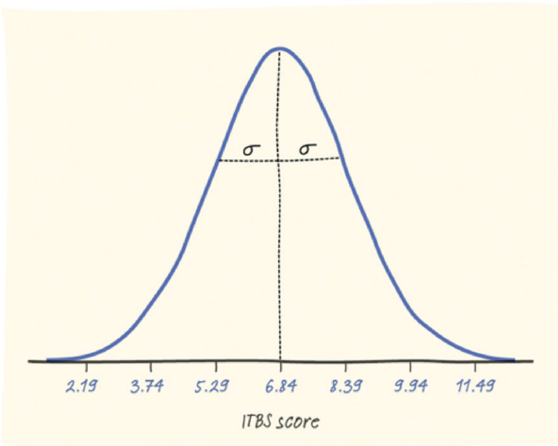
- Q. What is the mean of the distribution?
  - Q. What is the standard deviation of the distribution?

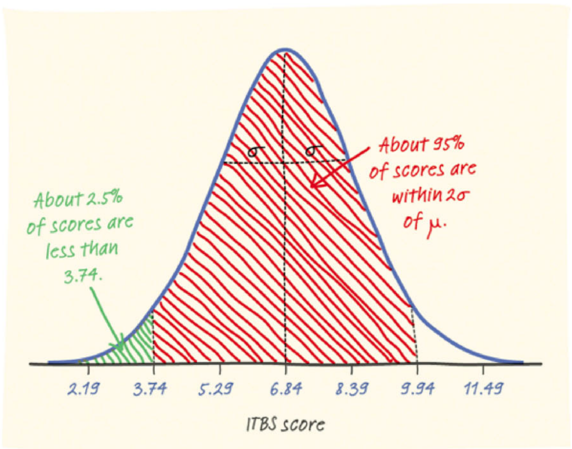




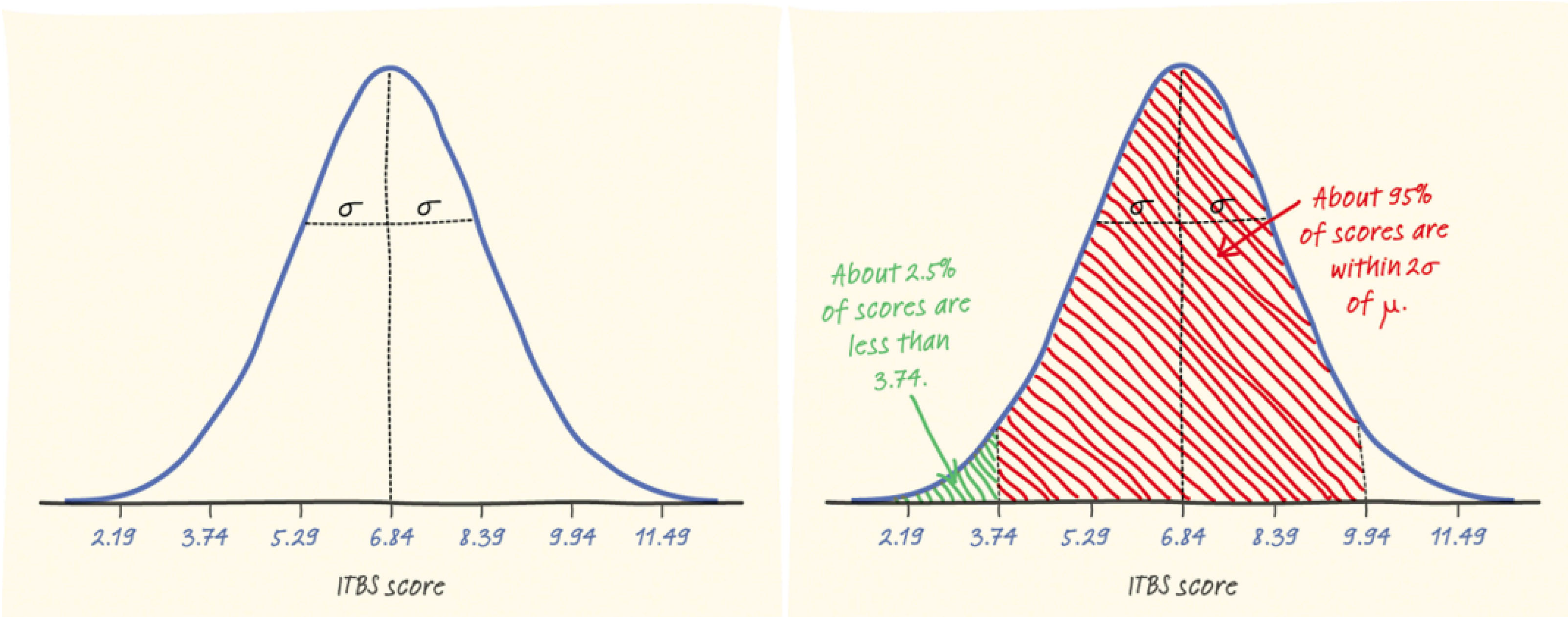
Worked Example

- The distribution of Iowa Test of Basic Skills (ITBS) vocabulary scores for seventh-grade students in Gary, Indiana, is close to Normal.
- Suppose the distribution is `N(6.84, 1.55)`.
- Sketch the Normal density curve for this distribution.
  - Q. What percent of ITBS scores is between 3.74 and 9.94?
  - Q. What percent of the scores is below 3.74?





Check your understanding: What percent of the scores is above 5.29?

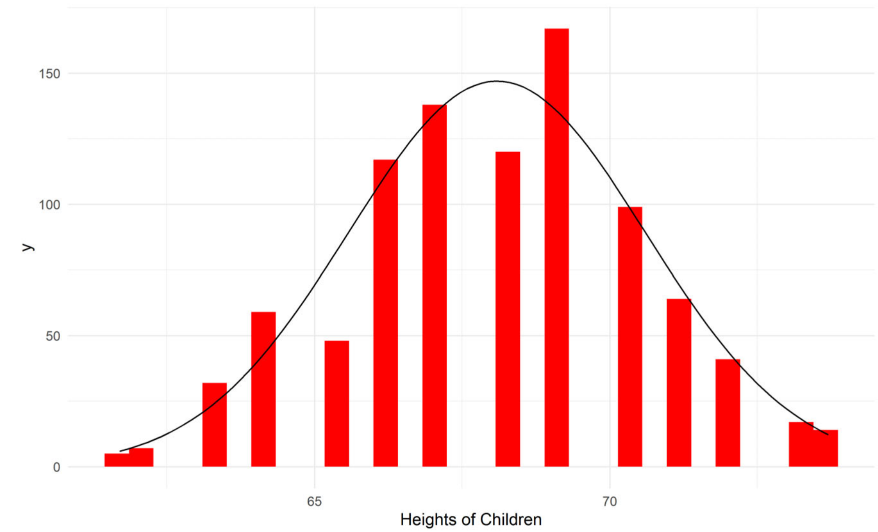


Standard Normal

- Normal is a model of the real world
- Not exact, but it is a facile model for many things
  - Physical features
  - Psychological features
  - Performance measures
  - Not all variables are normal
    - Skewed variables (e.g. income)
    - Any count variable (number of kids, mistakes on an exam)

Real World Data

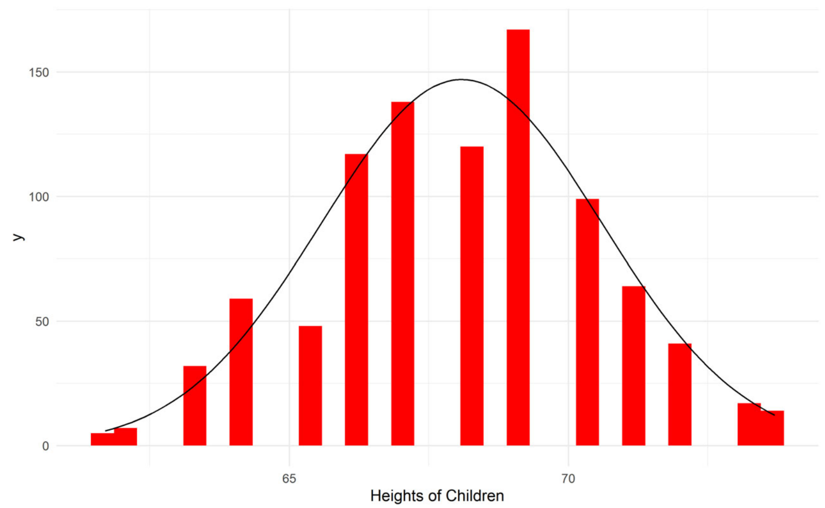
- Many variables follow this distribution (but not all)
- I have plotted histograms of data,
  - we have already used in this class
  - overlaid with the standard normal.



Methods in Psychological Research

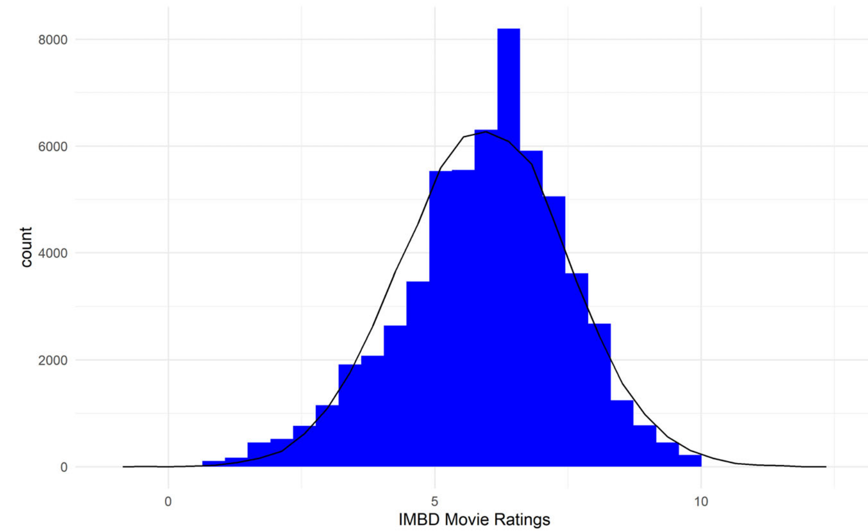
Height of Children (Galton dataset)

```
library(HistData)
library(ggplot2)
library(tidyverse)
Galton %>%
ggplot(aes(x = child)) +
    geom_histogram(fill = "red") +
    stat_function(
        fun = function(x, mean, sd, n){
            n * dnorm(x = x, mean = mean, sd = sd)
        },
        args = with(Galton, c(mean = mean(child), sd = sd(child), n
= length(child)))
    ) +
    scale_x_continuous("Heights of Children") + theme_minimal()
```

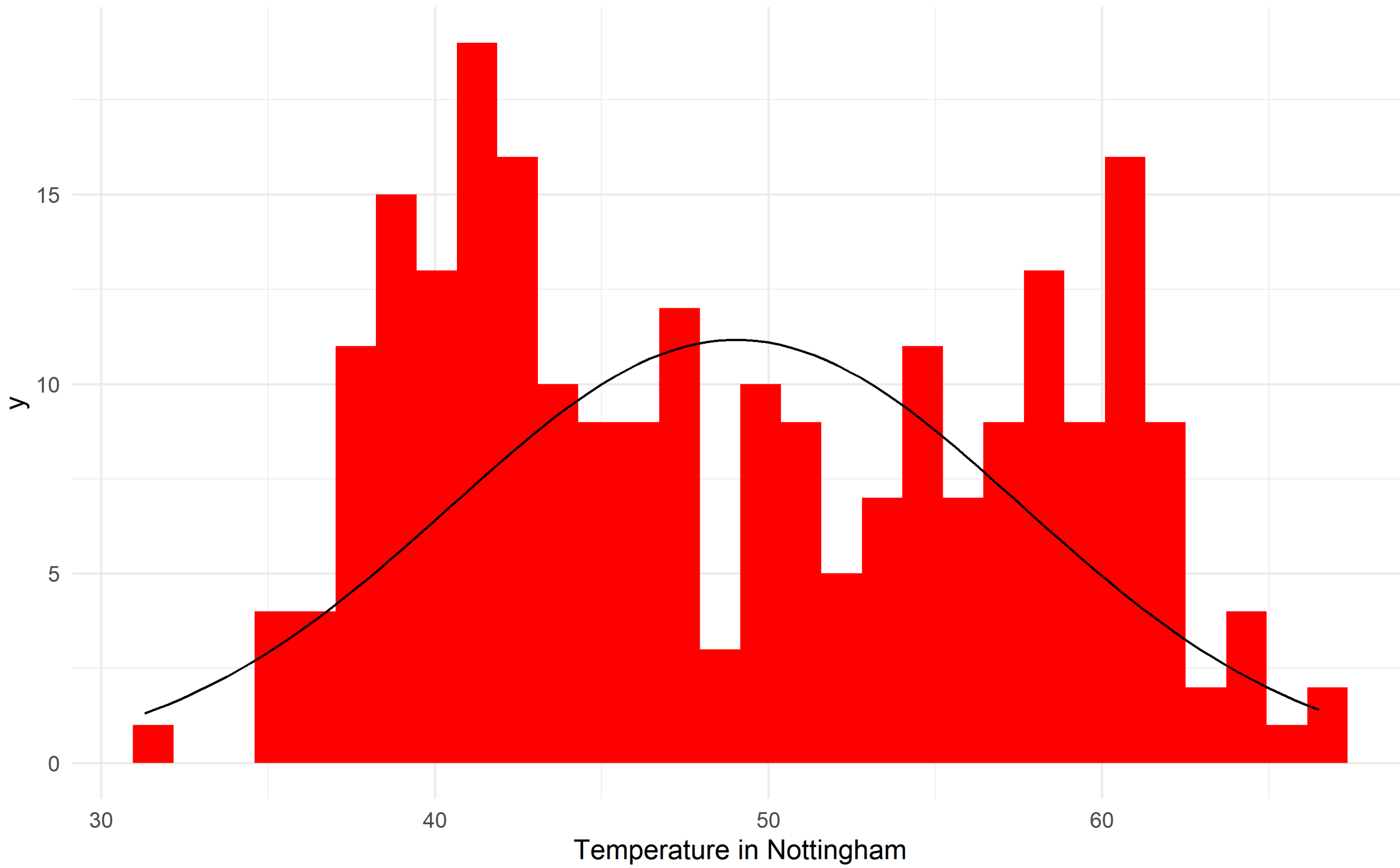


IMBD movie ratings (movies dataset)

```
library(ggplot2movies)
data(movies)

ggmovie <- ggplot(movies,
               aes(x = rating)) +
    geom_histogram(fill = "blue") +
    geom_freqpoly(aes(
      x = rnorm(length(rating))*sd(rating) + mean(ratin
                  fill = "black")  +
    scale_x_continuous("IMBD Movie Ratings") +
  theme_minimal()
```



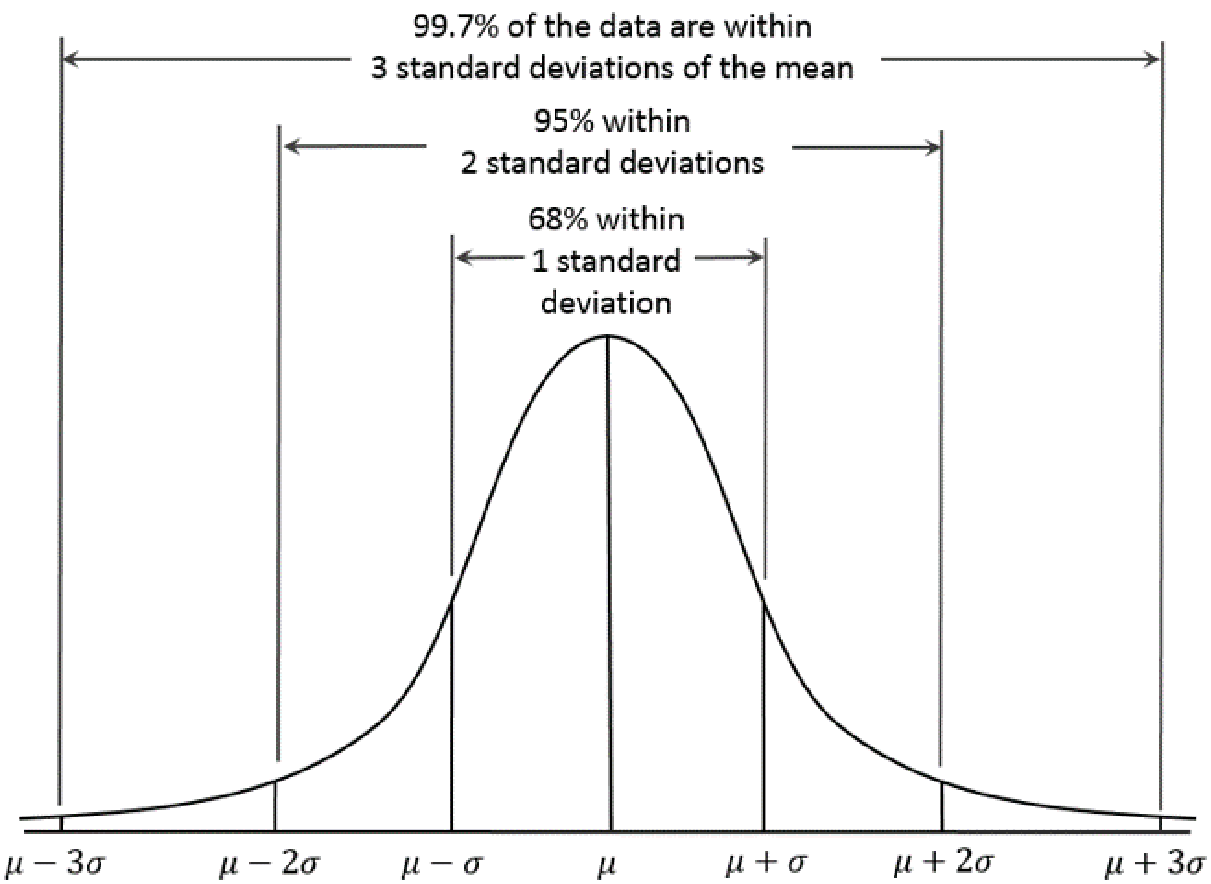
Temperature in Nottingham (nottem dataset)



Standard Normal

- Normal distribution tricks
  - Symmetric
  - 50% of area above zero
  - Total proportion is 1.0 (or 100%)

Area under the Normal Distribution



Wrapping Up...

Rescaling

Rescaling

- All Normal distributions are the same
  - if we measure in units of size σ from the mean μ as center.
- We can convert any variable into the same metric as the standard normal
- Changing to these units is called standardizing or rescaling.

Z-Score

- Z-score describes the location of the raw score in terms of distance from the mean, measured in standard deviations

- Statistics Sample
- $z_i = \frac{x_i - \bar{x}}{s}$

- Population
- $z_i = \frac{x_i - \mu}{\sigma}$

Merits

- Advantages
  - Allows us to compare scores on a common metric
  - Origin is 0. The mean
    - The units are 1, the standard deviation
    - '+' values above the mean
    - '-' values below the mean
- We can compare across measurement scales
  - Shape of the distribution does NOT CHANGE
- We can go from z-scores to raw scores

Demo

```
library(ggplot2movies)

# Raw data
movies$rating[1:10]
```

```
##  [1] 6.4 6.0 8.2 8.2 3.4 4.3 5.3 6.7 6.6 6.0
```

```
# Rescaling
variable <- movies$rating
scale(variable)[1:10]
```

```
##  [1]  0.30079877  0.04323788  1.45982279  1.45982279 -1.63090793
##  [6] -1.05139592 -0.40749368  0.49396944  0.42957922  0.04323788
```

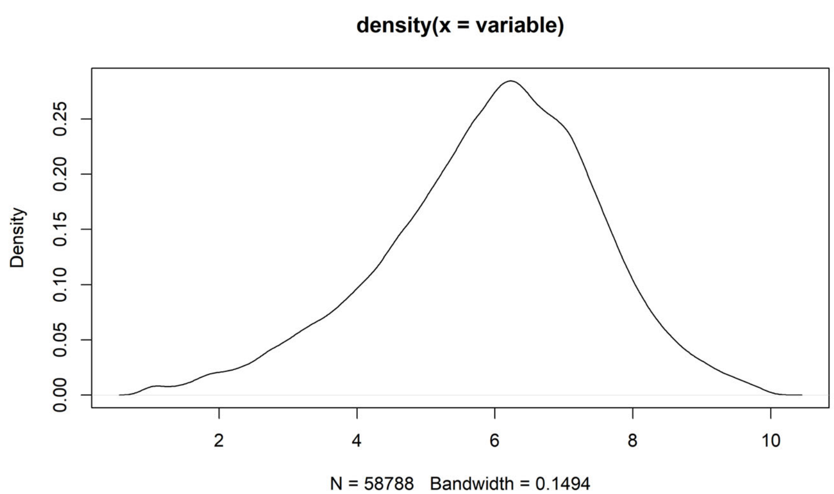
- Mean = 5.93 (SD = 1.55)

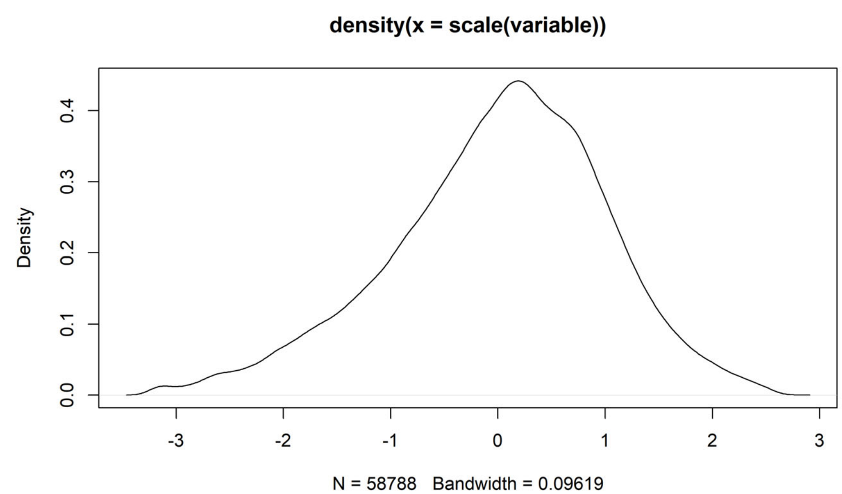
Raw	Z_Score
6.4	0.30
6.0	0.04
8.2	1.46
8.2	1.46
3.4	-1.63
4.3	-1.05
5.3	-0.41
6.7	0.49
6.6	0.43
6.0	0.04

Demo

```
plot(density(variable)) # no scalin
```

```
plot(density(scale(variable))) # wi
```





Z-Score

- Gives us information about the location of that score relative to the "average" deviation of all scores
- A z-score is the number of standard deviations a score is above or below the mean of the scores in a distribution
- A raw score is a regular score before it has been converted into a Z score
- Raw scores on very different variables can be converted into Z scores and directly compared

Worked Z-Score Problem

- Here are the IQ test scores of 31 7th-grade girls in a Midwest school district.

114	130	74	100	120	112	104	132
107	89	111	103	102	128	98	91
118	96	114	119	112	114	86	112
103	72	93	105	111	108	103	

Worked Z-Score Problem

A) We expect IQ scores to be approximately Normal.

- Make a stem plot to check that there are no major departures from normality.

13	2 0
12	8 0
11	9 8 4 4 4 2 2 2 1 1
10	8 7 5 4 3 3 3 2 0
9	8 6 3 1
8	9 6 4
7	4 2

Worked Z-Score Problem

B) Find the mean and standard deviation

- Mean =105.84 = $\sum \frac{X_i}{n}$ = 3281/31
- SD = 14.27 = s^2 = $\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$ = s^2 = $\frac{\sum_{i=1}^{n}(x_i - 105.84)^2}{30}$

Worked Z-Score Problem

C) What proportion of scores are within one standard deviation of the mean?

- One SD above mean = 105.84 + 14.27 = 120.11
- One SD below mean = 105.84 - 14.27 = 91.57
- 23/31 = 0.74

13	2 0
12	8 0
11	9 8 4 4 4 2 2 2 1 1
10	8 7 5 4 3 3 3 2 0
9	8 6 3 1
8	9 6 4
7	4 2

Worked Z-score Problem

B) What proportion of scores are within TWO standard deviations of the mean?

+ TWO SD above mean = 105.84 + 2*(14.27) = 134.38
+ TWO SD below mean = 105.84 - 2*(14.27) = 77.3
+ 29/31 = 0.935

Worked Z-score Problem

B) What would these proportions be in an exactly Normal distribution?

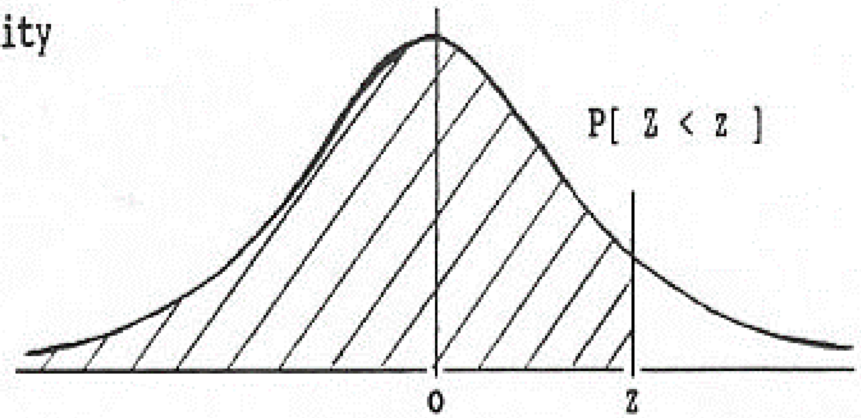
- +/- One SD?

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability up to the standardised normal value z i.e.

$$P[\, Z < z \,] = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp(-\tfrac{1}{2}Z^2)\, dZ$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09

Moving to powerpoint